Decoupling-Oriented Coordination of Paralleled VSGs With Multi-Agent Reinforcement Learning

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***Abstract*—Within the learned-policy canary bank, adaptive virtual inertia and damping can satisfy decoupling endpoints while violating comparator-relative action limits. This paper evaluates that distinction in a modified Kundur system with four virtual synchronous generators (VSGs) after correcting the device-to-system-base conversion. Four independent soft actor–critic agents provide one way to adapt virtual inertia M and damping D. We define a complete guard-first contract that separates common-frequency support, off-diagonal and differential response energies, and physical and action no-harm checks. A deterministic direct-M/D law selected on two development profiles passes the predefined deterministic guard set on four prospectively fresh profiles at both 6 s and 30 s. We then audit 208 frozen policies from a prospectively trained $2 \times 2 \times 2$ actor-source, critic-source, and reward-access factorial with 26 paired training seeds. On a separate four-profile 30 s canary bank, 126 policies meet both aggregate 5% endpoint-improvement targets relative to direct M/D. None passes the complete contract: all 832 policy–profile blocks violate both registered relative action-RMS and action-variation limits. At the separately reported 6 s and 30 s horizons, none of four source-factor contrasts establishes improvement above the 10% materiality boundary after Holm control. For these separate deterministic-fresh and learned-canary banks under the registered parameter card, endpoint-only assessment qualifies policies that the complete contract rejects; this does not imply a universal failure of multi-agent reinforcement learning (MARL).**

***Keywords*—virtual synchronous generator, multi-agent reinforcement learning, virtual inertia, frequency coordination, guard-first evaluation**

I. Introduction

Virtual synchronous generators (VSGs) reproduce swing-like behaviour in converter controls and expose tunable inertia and damping parameters [1]–[3]. When several units operate in parallel, fleet-average support and inter-unit motion are different physical coordinates. Model-based consensus, mutual damping, and parallel-VSM designs make that distinction explicit through their feedback structure [4]–[9].

Relevant precedents include distributed dynamic inertia–droop MARL for paralleled VSGs [10], single-VSG soft actor–critic M/D tuning [11], MADDPG joint-M/D control [12], decentralized multi-VSG MARL [13], and constraint-aware power-system RL [14], [15]. We inherit their broad controller and learning elements. Our contribution is instead the post-correction audit of a frozen family using separated development, fresh, and canary banks, matched source interventions, split horizons, and comparator-relative guards. It is neither a new learning algorithm nor a basis for cross-system inference. However, an aggregate reward or average-frequency trajectory cannot by itself identify differential coordination. A favourable endpoint can also coexist with large or rapidly varying commands, while actor, critic, and reward information may change simultaneously in a nominal message/no-message comparison. Reinforcement-learning seed variability further limits small-sample conclusions [16].

These issues became decisive in the present study. An audit found that the original controller arithmetic and simulator runtime used different M/D bases. After applying the device/system-base conversion exactly once, the previous positive evidence was no longer admissible. A separate zero-adaptive-action sensitivity bank also showed that a 6 s window can miss the tail at high inertia. Accordingly, this paper treats unit correctness, horizon separation, and the complete contract as part of the scientific question rather than as implementation details.

Our goal is to determine whether corrected M/D-adaptive MARL provides decoupling gains beyond a development-selected deterministic reference while satisfying every registered guard. We use two signed finite-window response energies, require per-profile validity and no-harm checks before interpreting endpoints, and reserve the fresh evaluation bank for the deterministic feasibility claim. Learned policies are judged on a separate canary bank, so evidence is not transferred between controllers or profile splits.

The evaluation combines a development-selected local-neighbour direct-M/D law with a prospectively trained $2 \times 2 \times 2$ source/reward factorial. Twenty-six paired seeds support actor, critic, actor-by-critic, and critic-by-reward contrasts. The registered 6 s primary analysis and the evaluation-only 30 s tail sensitivity share frozen policies and are therefore reported side by side but never pooled. Figure 1 summarizes the two decision paths.

This paper makes four bounded contributions. First, Section II specifies a corrected device/system-base action contract and separates common and differential coordinates from empirical guard decisions. Second, Sections III–IV establish a constructive deterministic reference that passes four prospectively fresh 30 s profiles. Third, Section IV reports a complete 208-policy tail audit showing that endpoint qualification and complete-contract satisfaction can diverge: 126 policies meet both endpoint targets, while none satisfies the complete

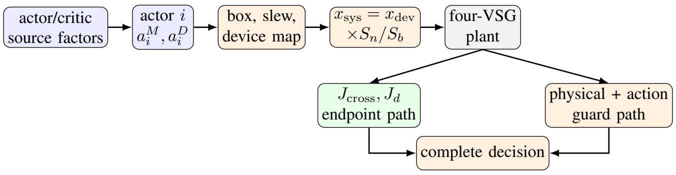

Fig. 1. Corrected guard-first evaluation. Per-VSG actions are decoded on the device base and converted exactly once at the simulator boundary. Endpoint improvement is interpreted only after the physical and action guard path passes.

contract. Fourth, Sections III–IV show that the matched 26-seed factorial does not establish improvement above the 10% materiality boundary for any registered source contrast after multiplicity control.

II. Coordination Object and Guard-First Contract

A. Corrected physical and action object

We consider four controlled VSGs in a modified Kundur two-area system. Their angle and frequency states follow

$$\dot{\delta}_i = 2\pi f_n(\omega_i - 1), \qquad M_i^{\mathrm{sys}}\dot{\omega}_i = \tau_{m,i} - \tau_{e,i} - D_i^{\mathrm{sys}}(\omega_i - 1), \tag{1}$$

where reported physical frequency uses the Kundur $f_n = 60\,\mathrm{Hz}$ base. Equation (1) fixes the physical state convention used below. Controller arithmetic is performed on the device base, whereas ANDES stores runtime parameters on the system base [17]. Each boundary crossing therefore applies exactly one conversion,

$$x_i^{\mathrm{sys}} = x_i^{\mathrm{dev}}\frac{S_{n,i}}{S_b}, \qquad x \in \{M, D\}, \qquad M = 2H. \tag{2}$$

The registered parameter card uses $S_b = 100\,\mathrm{MVA}$, $S_{n,i} = 200\,\mathrm{MVA}$, $H_0^{\mathrm{dev}} = 100\,\mathrm{s}$, $M_0^{\mathrm{dev}} = 200\,\mathrm{s}$, and $D_0^{\mathrm{dev}} = 100$. These values define the corrected project calibration; they do not reconstruct an unspecified baseline from prior literature.

At each $0.2\,\mathrm{s}$ control instant, controller i produces a normalized target $\bar{a}_i$. Let $\Pi_c(x) = \min\{c, \max\{-c, x\}\}$ componentwise. With $a_i^{\mathrm{exec}}[-1] = 0$, the common projection is

$$\begin{aligned} a_i^{\mathrm{amp}}[k] &= \Pi_1(\bar{a}_i[k]), \\ a_i^{\mathrm{exec}}[k] &= \Pi_1(a_i^{\mathrm{exec}}[k-1] \\ &\qquad + \Pi_{0.25}(a_i^{\mathrm{amp}}[k] - a_i^{\mathrm{exec}}[k-1])). \end{aligned} \tag{3}$$

The projected components then enter the decoder

$$\begin{aligned} q(a) &= \begin{cases} 600a, & a \geq 0, \\ 200a, & a < 0, \end{cases} \\ M_i^{\mathrm{dev}}[k] &= \max\{20, M_{i,0}^{\mathrm{dev}} + q(a_i^{M,\mathrm{exec}}[k])\}, \\ D_i^{\mathrm{dev}}[k] &= \max\{10, D_{i,0}^{\mathrm{dev}} + q(a_i^{D,\mathrm{exec}}[k])\}. \end{aligned} \tag{4}$$

After (2), each decoded target is linearly applied over five simulator substeps from the preceding runtime readback. Thus, projection precedes decoding; no learned action directly injects active power.

B. Common–differential endpoints

Let $\Delta f \in \mathbb{R}^4$ be the physical VSG-frequency deviations. We use $z_c = \mathbf{1}^\top \Delta f/4$ and

$$z_d = T_d \Delta f, \quad T_d = \begin{bmatrix} 1/2 & 1/2 & -1/2 & -1/2 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \\ 0 & 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}. \tag{5}$$

The coordinates in (5) are registered arithmetic coordinates, not identified modal coordinates, and z_c is not an inertia-weighted centre-of-inertia frequency.

For probe type $k \in \{c, d, \ell\}$, let $\varepsilon_c = \varepsilon_d = q$ and $\varepsilon_\ell = \ell$ for that profile. The odd response of a matched signed pair is $\widetilde{f}_k[n] = (\Delta f_k^+[n] - \Delta f_k^-[n])/2$. With $z_{c,k} = \mathbf{1}^\top \widetilde{f}_k/4$ and $z_{d,k} = T_d \widetilde{f}_k$, the lower-is-better endpoints are

$$J_{\mathrm{cross}} = \frac{\Delta t}{\varepsilon_c^2}\sum_{n=1}^{N_T}\frac{\|z_{d,c}[n]\|_2^2}{3} + \frac{\Delta t}{\varepsilon_d^2}\sum_{n=1}^{N_T} z_{c,d}^2[n], \tag{6}$$

$$J_d = \sum_{k\in\{c,d,\ell\}}\frac{\Delta t}{\varepsilon_k^2}\sum_{n=1}^{N_T}\frac{\|z_{d,k}[n]\|_2^2}{3}. \tag{7}$$

The endpoints in (6) and (7) use $\Delta t = 0.2\,\mathrm{s}$ and $N_T = T/\Delta t$ for the separately reported $T = 6\,\mathrm{s}$ and $T = 30\,\mathrm{s}$ horizons. The quantities are discrete response energies, not joules, storage functions, or Lyapunov functions.

C. Guard-first decisions

The following checks define the *complete guard-first contract*, shortened to *complete contract* below. Endpoint improvement is interpreted only after trajectory validity, actuator mapping, normalized box and slew limits, common-frequency integral, worst-unit peak, RoCoF, command stress, saturation, and action-nondegeneracy checks. For learned policies, the three frequency quantities must be no worse than 103% of the same-profile deterministic reference; executed-action RMS and total variation must be no worse than 110%; saturation fraction must be at most 0.05; and the registered variation and per-VSG-dispersion floors are 10^{-6}. These limits were registered before the canary outcomes and imported by hash. They are empirical comparator-relative tolerances to prevent endpoint gains from masking frequency (3%) or command-stress (10%) degradation, not industry- or hardware-calibrated limits. Because the sealed audit stores qualification rather than action-ratio margins, rejection is limited to these exact thresholds; neighbouring-cutoff sensitivity is unassessed.

A learned policy passes the complete contract only when every one of four canary profiles passes all per-profile guards and the equal-weight aggregate ratios of both J_{cross} and J_d are at most 0.95 relative to the deterministic controller. Deterministic fresh-bank feasibility uses a predefined zero-action-referenced version of these checks, including 3% frequency no-harm limits and the same mapping, bound, slew, saturation, nonconstant-action, and per-VSG-dispersion checks. Its endpoint ratios are descriptive report lines rather than gate inputs. Training reward cannot override either decision. This follows the broader separation between optimization penalties and verified closed-loop constraints in safe power-system reinforcement learning [14], [15]. These empirical guards are not stability or safety certificates.

III. Corrected Adaptive MARL Evaluation

A. Profiles and deterministic reference

Each profile contains common, differential, and localized probes, each with positive and negative signs, giving six signed scenarios. The frozen learner retains its registered 50 Hz controller scaling, whereas physical frequency endpoints use the 60 Hz plant base. The parameter-card anchor is $H_0 = 100\,\mathrm{s}$ unless overridden by the registered profile vectors in Table I. For a profile with magnitudes q and ℓ, the common probe applies $\pm q/4$ at each of four loads, the differential probe applies $\pm q/4$ to PQ0/PQ1 and $\mp q/4$ to B14/B15, and the localized probe applies $\pm \ell$ at the listed load. Table I gives every profile used for selection, training, or evaluation.

Learner episodes cycle deterministically through the 24 T1–T4 signed profile–scenario combinations. F1–F4 are reserved for deterministic evaluation, and C1–C4 are reserved for learned-policy evaluation.

A separate zero-action screen showed that the worst-unit peak absolute frequency deviation over 6 s changed by 89.8%–92.9% at $H_0 = 10\,\mathrm{s}$ and by -34.6%–-34.3% at $H_0 = 300\,\mathrm{s}$ relative to the anchor; at high inertia this peak could occur after 6 s. We therefore add a separate 30 s tail without claiming cross-H controller robustness.

The deterministic family contains zero action and nine local-neighbour direct-M/D laws with $(k_M, k_D) \in \{0.5, 1, 2\}^2$. Let $\mathcal{N}_i$ contain agent i's two ring neighbours. The direct law uses dimensionless, physical-60 Hz-calibrated features $\phi_i = 2\pi 60(\omega_i - 1)/3$ and $\rho_i = 2\pi 60\dot{\omega}_i/5$; equivalently, it multiplies the registered learner observation's six frequency/RoCoF slots by 60/50, while learned policies retain the registered 50 Hz scaling. Before the common projection and decoder, each direct law forms

$$\begin{aligned}
\sigma_i &= |\phi_i| + |\rho_i|, \\
\bar{a}_i^M &= \tanh\left[k_M\left(\sigma_i - \frac{1}{2}\sum_{j\in\mathcal{N}_i}\sigma_j\right)\right], \\
\bar{a}_i^D &= \tanh\left[k_D\left(|\phi_i| + \frac{1}{2}\sum_{j\in\mathcal{N}_i}(|\phi_i - \phi_j| + |\rho_i - \rho_j|)\right)\right].
\end{aligned} \tag{8}$$

A candidate is development-eligible only if all runs are valid, common-frequency no-harm checks pass, and saturation is at most 0.05. Among eligible candidates, a frozen ascending lexicographic rule minimizes the worse of the two D1–D2 aggregate endpoint ratios to zero action, their sum, summed action RMS, and finally the arm-identifier string as an exact tie-breaker. It selected $(k_M, k_D) = (2, 2)$. We then evaluated the selected law on F1–F4 without reselection. The same controller and profiles were rerun to 30 s; the 6 s and 30 s observations are not counted as eight independent profiles.

B. Prospectively trained source factorial

The MARL family contains four independent per-VSG soft actor–critic learners [18]. It is neither parameter-shared nor a joint-critic centralized-training/decentralized-execution method such as MADDPG [19]. Each local state contains active power, frequency, RoCoF, two ring-neighbour frequency/RoCoF pairs, and the agent's previously executed two-component action. Specifically, $o_i = [P_i/2, \delta\omega_i/3, \dot{\delta\omega}_i/5, \delta\omega_{j_1}/3, \delta\omega_{j_2}/3, \dot{\delta\omega}_{j_1}/5, \dot{\delta\omega}_{j_2}/5]$ and $s_i = [o_i, a_{i,t-1}^{\mathrm{exec}}] \in \mathbb{R}^9$. The Gaussian–tanh actor and twin-Q critic each use four 128-unit ReLU hidden layers. Adam uses a learning rate of 3×10^{-4}, with $\gamma = 0.99$, target-update rate $\tau = 0.005$, replay capacity 10,000, batch size 256, target entropy -2, automatic entropy coefficient in $[0.005, 5]$, and unit gradient-norm clipping. Once replay holds 256 transitions, one update follows each environment step. Projected executed actions enter the current, target, and actor Q paths.

Actor and critic observation views are intervened separately. Source N is the authentic same-time neighbour tuple. For the frozen circular donor map $\pi(i) = (i+1) \bmod 4$, placebo source P keeps the three recipient-local entries and takes the four neighbour entries from donor row $\pi(i)$: $P_i = [N_{i0}, N_{i1}, N_{i2}, N_{\pi(i)3}, N_{\pi(i)4}, N_{\pi(i)5}, N_{\pi(i)6}]$. Thus, recipients 0–3 use donor rows 1, 2, 3, 0. Thus, N versus P is a same-time total source intervention, not an isolated semantic-information effect. The per-agent training reward is

$$r_i = 100 r_{f,i} + 50 r_{\mathrm{abs},i} + 0.0056 r_H + 0.0056 r_D, \tag{9}$$

where reward access $r \in \{0, 1\}$, $\Delta f_i^{\mathrm{Hz}} = 3o_i[1]/(2\pi)$, $\overline{\Delta f}_i = (\Delta f_i^{\mathrm{Hz}} + r\sum_{j\in\mathcal{N}_i}\Delta f_j^{\mathrm{Hz}})/(1+2r)$, and

$$\begin{aligned}
r_{f,i} &= -(\Delta f_i^{\mathrm{Hz}} - \overline{\Delta f}_i)^2 - r\sum_{j\in\mathcal{N}_i}(\Delta f_j^{\mathrm{Hz}} - \overline{\Delta f}_i)^2, \\
r_{\mathrm{abs},i} &= -(\Delta f_i^{\mathrm{Hz}})^2, \\
r_H &= -\left(\frac{\frac{1}{4}\sum_m \Delta M_m}{600}\right)^2, \quad r_D = -\left(\frac{\frac{1}{4}\sum_m \Delta D_m}{600}\right)^2.
\end{aligned} \tag{10}$$

Reward access switches only the neighbour-dependent terms; reward is never used by the evaluation contract.

The experiment crosses actor source $\{N, P\}$, critic source $\{N, P\}$, and reward access $\{0, 1\}$; the last factor switches only the registered neighbour-dependent reward terms. Twenty-six matched seeds, 501–526, populate all eight arms for 208 newly trained cells. Policies from earlier pilot training do not enter the inference. Each cell could run from 30,000 to 43,200 interaction steps, with checks comparing adjacent 2,000-update windows and termination allowed only after three consecutive qualifying checks. A check requires finite loss, entropy-coefficient, and actor-gradient histories. It also requires actor- and critic-loss median absolute magnitudes within a factor 1.10 across adjacent windows; entropy-coefficient medians either both at most 1.01(0.005) or within a factor 1.05; an actor-gradient median absolute magnitude of at least 10^{-6} and window ratio within a factor 1.25; zero time-domain-simulation failures; and at most 2% drift on a fixed-state

TABLE I
REGISTERED PROFILE CARDS. M_0, D_0 ARE FOUR-VSG DEVICE-BASE VECTORS; P_{14}, P_{15}, q, ℓ ARE SYSTEM-P.U. ACTIVE-POWER QUANTITIES. D DENOTES DETERMINISTIC DEVELOPMENT, T LEARNER TRAINING, F DETERMINISTIC FRESH EVALUATION, AND C LEARNED-POLICY CANARY EVALUATION.

ID	Role	M_0	D_0	P_{14}	P_{15}	q	ℓ	Local load
D1	development	[230,220,190,250]	[70,80,150,110]	1.92	0.42	0.70	1.05	B14
D2	development	[180,140,150,210]	[140,100,70,50]	2.08	0.52	1.15	0.95	B15
T1	training	[150,250,170,230]	[60,140,80,120]	2.24	0.42	0.85	0.95	PQ1
T2	training	[230,150,250,170]	[120,60,140,80]	2.02	0.66	1.05	1.15	B15
T3	training	[210,190,160,240]	[130,70,110,90]	2.42	0.14	0.75	0.85	PQ0
T4	training	[240,160,190,210]	[90,110,70,130]	2.12	0.54	0.95	1.05	B14
F1	fresh eval.	[190,200,140,160]	[110,120,130,70]	1.92	0.64	1.15	0.95	PQ1
F2	fresh eval.	[230,140,250,150]	[60,90,70,110]	2.04	0.58	1.00	1.20	PQ0
F3	fresh eval.	[200,170,250,140]	[60,80,70,140]	2.40	0.36	0.90	0.85	B14
F4	fresh eval.	[190,150,170,220]	[120,130,80,110]	2.58	0.28	0.75	0.90	PQ0
C1	canary eval.	[140,260,200,220]	[50,150,90,130]	2.56	0.34	0.90	1.00	PQ0
C2	canary eval.	[260,140,220,200]	[150,50,130,90]	2.06	0.26	0.80	0.90	B14
C3	canary eval.	[180,240,150,210]	[70,130,60,110]	1.96	0.64	1.00	1.10	B15
C4	canary eval.	[220,200,260,140]	[110,90,150,50]	2.32	0.46	1.10	1.20	PQ1

deterministic-action probe. The fixed 21,600-step midpoint and stopping-rule final checkpoints define 16 registered arm-by-checkpoint evaluation batches; the final checkpoint and 6 s J_d form the primary factorial.

For statistical reconstruction, let $y_{acrps} = \log L_{acrps}$, where L is the positive final-checkpoint J_d for actor source a, critic source c, reward access r, profile p, and seed s. With angle brackets denoting equal weighting over the displayed nuisance indices, the four within-seed estimands are

$$\begin{aligned} d_s^A &= \langle y_{Pcrps} - y_{Ncrps} \rangle_{c,r,p}, \\ d_s^C &= \langle y_{aPrps} - y_{aNrps} \rangle_{a,r,p}, \\ d_s^{AC} &= \langle y_{PNrps} - y_{NNrps} - y_{PPrps} + y_{NPrps} \rangle_{r,p}, \\ d_s^{CR} &= \langle y_{aP1ps} - y_{aN1ps} - y_{aP0ps} + y_{aN0ps} \rangle_{a,p}. \end{aligned} \tag{11}$$

Thus, positive main effects mean lower loss under authentic source N; positive interactions follow the registered ratio-of-ratios direction. The 26 matched seeds, not profiles or trajectories, are the inferential units. Tests centre (11) at the materiality boundary $\log(1.10)$. The exact upper-tail Wilcoxon test is primary when centred differences contain neither zeros nor tied absolute ranks and have absolute Fisher–Pearson skewness at most 1; otherwise, a prespecified one-million-draw upper-tail sign-flip mean test (seed 20260825) is primary. All four observed contrasts passed the exact route; sign flipping remains a sensitivity. Holm control covers the four-test family [20]. Table II descriptively reports $\Delta_{\text{geo}} = 100\{\exp(\langle d_s \rangle_s) - 1\}\%$, positive toward an authentic-source benefit. The signed-rank test targets a rank-based location shift, not this mean-log summary; its p values are not an interval for Δ_{geo}.

The tail audit freezes all 208 final policies and introduces no training, tuning, checkpoint selection, or parameter change. Policies run for 150 steps on four registered canary profiles; learned policies never run on the fresh bank. The complete inventory contains 16/16 valid shards, 848/848 profile blocks (832 learned-policy and 16 zero/direct-M/D reference blocks), and 5,088/5,088 valid trajectories. The 30 s source calculation uses the same 26 paired seeds and is a sensitivity analysis, not an independent replication of the 6 s primary analysis.

IV. RESULTS

Figures 2 and 3 report, respectively, deterministic fresh-bank feasibility and learned canary-bank endpoint and contract decisions; the evidence objects remain non-pooled.

A. Deterministic reference feasibility

The corrected mapping and trajectory-integrity checks pass. The selected direct-M/D controller also passes the predefined deterministic guard on all four prospectively fresh profiles at 6 s and, separately, at 30 s. At the latter horizon, the off-diagonal ratios to zero action range from 0.3987 to 0.5559, and the differential ratios range from 0.3307 to 0.5580. The plotted

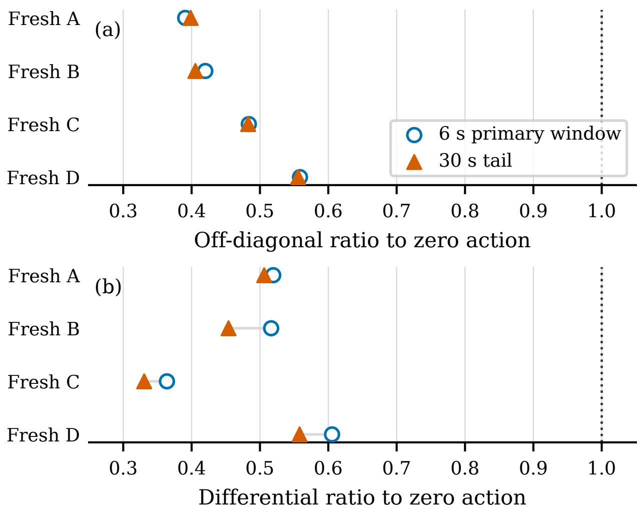

Fig. 2. The development-selected direct-M/D law remains guard-valid on all four prospectively fresh profiles at the registered 6 s window and the separate 30 s tail. Lines connect the same profile across horizons; the dotted line is zero-action parity. Ratios are descriptive report lines, not deterministic-guard inputs.

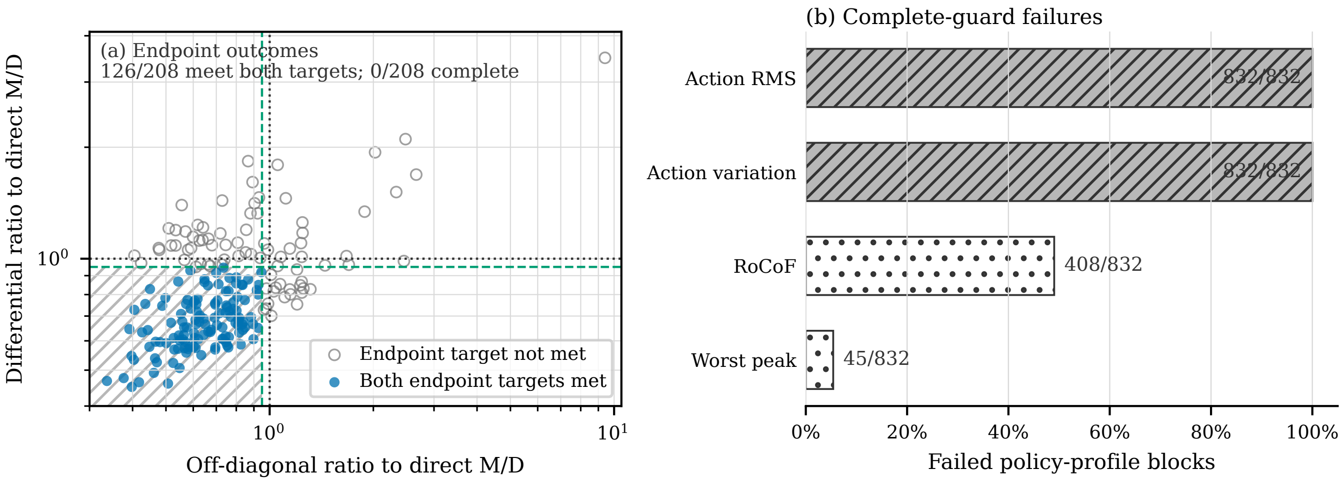

Fig. 3. Endpoint improvement is not complete-contract satisfaction. (a) Each point is one frozen policy on the equal-weight four-profile canary aggregate; lower is better. Green dashed lines mark the two 0.95 endpoint targets, black dotted lines mark parity, and hatching marks their joint target region. (b) Exact failure counts across all 832 policy–profile blocks. Although 126/208 policies meet both aggregate endpoint targets, 0/208 passes the complete contract because every block violates both comparator-relative action-stress limits.

ratios describe response reduction; they are not deterministic-guard inputs.

Direct M/D also passes 4/4 canary profiles, but that result is descriptive and used only to validate the learned-policy reference. It is not pooled with the fresh result.

B. Endpoint improvement does not satisfy the complete contract

The learned-policy audit yields a sharp endpoint/action-limit separation. Although 126/208 policies reduce both equal-weight aggregate endpoints by at least 5% relative to direct M/D, no policy passes the complete contract and no policy passes the full per-profile guard set on even one canary profile. All 832/832 learned policy–profile blocks violate both registered relative action-stress limits. Additionally, 408 blocks fail RoCoF no-harm and 45 fail worst-unit-peak no-harm, while all common-frequency-integral checks pass.

The result is therefore not an absence of endpoint movement. Rather, the endpoint reductions coexist with command stress, and for some blocks with frequency stress, beyond the frozen comparator-relative allowances. Those relative limits are not absolute hardware-energy bounds.

C. Registered source effects

Every one of the 208 training cells reached 43,200 steps without satisfying the registered convergence certificate. Adaptive stopping therefore saved no training time, but this neither invalidates the completed evaluations nor proves divergence. Table II reports the four contrasts at the two non-pooled horizons. No contrast establishes improvement above the 10% materiality boundary after Holm control.

The actor-by-critic point estimates cross the 10% materiality boundary, but their raw one-sided p values of 0.02968 and 0.02037 become 0.1187 and 0.0815 after Holm correction.

TABLE II
NON-POOLED DESCRIPTIVE SOURCE ESTIMATES AND HOLM-ADJUSTED TESTS

Effect	6 s primary Δ_{geo}	6 s primary p_{Holm}	30 s sensitivity Δ_{geo}	30 s sensitivity p_{Holm}
Actor main	-3.50%	1.0000	-4.99%	1.0000
Critic main	$+5.20\%$	1.0000	$+9.67\%$	1.0000
Actor×critic	$+21.81\%$	0.1187	$+28.15\%$	0.0815
Critic×reward	$+4.56\%$	1.0000	-5.37%	1.0000

They are therefore hypothesis-generating rather than confirmed material interactions. Failure to establish a material interaction does not establish a zero effect or equivalence.

V. Discussion and Limitations

The deterministic result shows that the corrected direct-M/D feasible set is nonempty within the registered finite bank. The learned result identifies a different boundary: endpoint-only assessment would qualify 126/208 policies, whereas the complete contract qualifies 0/208. The central lesson is not that the tested learners cannot reduce differential response, but that the policy-level endpoint and block-level action decisions separate: 126 policies meet both aggregate endpoint targets, while all 832 blocks violate both frozen relative action allowances. This establishes tested-bank co-occurrence, not a causal exchange between endpoint reduction and stress for each qualified policy.

Earlier RL-based VSG studies use different plants, learners, rewards, and validation objectives [10]–[13]. The combined unit-corrected, bank- and horizon-separated, guard-first audit is the bounded distinction here. Its findings neither confirm nor contradict their controller improvements; they delimit this tested contract.

The source factorial adds a second boundary. Neither horizon establishes a registered source effect above 10% after multiplicity control. Moreover, the N/P intervention can combine source authenticity, optimization, contemporaneous dependence, and distribution shift; it does not isolate pure semantic neighbour-information value.

The conclusions cover exactly 208 frozen policies, four canary profiles, six signed scenarios per profile, one Kundur topology under the corrected parameter card, and the registered learner family. Learned policies were not evaluated on the fresh bank; the fresh 4/4 result belongs only to the deterministic controller. The 6 s primary and 30 s sensitivity share policies and seeds and are not replications. Similarly, canary and fresh deterministic banks are not pooled.

Finally, the study provides no population failure probability, cross-H controller robustness, topology generalisation, stability or safety certificate, EMT or HIL validation, or deployment claim. Communication is synchronous and idealized; delay, loss, and quantization remain outside the contract. Claims about learned-policy performance on fresh profiles, robustness across topologies or H, or hardware performance would require separate prospective experiments. An evidence ladder would perturb delay, loss, and quantization against source inputs and command-variation guards, use electromagnetic-transient (EMT) simulation to challenge converter dynamics and M/D mapping, then use hardware-in-the-loop (HIL) tests for timing, measurement, and actuation effects.

The preserved archive binds reviewed commits and SHA-256 source identities, 208 checkpoints, parameter/profile cards, 16 shards with 5,088 trajectories, and regeneration scripts; its runtime record is Python 3.12.3 and ANDES 2.0.0. No public archive identifier, complete dependency lock, or container is yet available, so this supports internal traceability rather than unrestricted replay. The source analysis supplies no interval compatible with its signed-rank estimand; precision beyond the adjusted decisions is unquantified.

VI. Conclusion

With the corrected device-to-system-base mapping, the selected deterministic direct-M/D controller passes all four fresh 30 s profiles. In contrast, none of 208 frozen MARL policies passes the complete contract on the canary bank: 126 satisfy both aggregate endpoint targets, but all 832 policy–profile blocks violate both registered action-stress limits. No registered source contrast establishes improvement above the 10% materiality boundary after Holm control at either the 6 s primary horizon or the separate 30 s sensitivity. For the finite family tested here, the complete contract therefore changes the policy-qualification decision suggested by endpoints alone, without implying a universal failure of M/D-adaptive MARL.

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